Part B — Baseline Unit Testing Suite Execution

Test Case ID	Target Function / Operation	Test Input Vector	Expected Output	Actual Output (Legacy Code)	Pass / Fail	Technical Remarks
TC 01	`add_item()`	10,20,30	Three items stored	Items added successfully.	**PASS**	Normal insertion works.
TC 02	`add_item()`	6 items	Capacity expands; all retained.	Items added but old array is leaked.	**FAIL**	Resize does not free old memory
TC 03	`printAll()`	5 items	Exactly 5 valid items.	Reads one extra element.	**FAIL**	<= causes out of-bounds access.
TC 04	`findItem()`	`Search 30`	Returns index 2.	Returns 2.	**PASS**	Search works despite redundant loop.
TC 05	`findItem()`	`Search 999`	Returns -1	Returns -1.	**PASS**	Missing target handled

TC 06	`remove_item_at()`	`Delete index 2`	Element removed and shifted.	Works correctly.	**PASS**	Functional but inefficient.
TC 07	`remove_item_at()`	`Delete -5`	Reject operation	Invalid index!	**PASS**	Negative index
TC 08	`remove_item_at()`	`Delete 99`	Reject safely.	Possible invalid memory access.	**FAIL**	`No upper-bound`
TC 09	`processMatrix()`	3×3 transpose	Correct transpose.	Correct transpose.	**PASS**	Works for fixed
TC 10	Memory Clean	Exit program	No memory leaks.	Memory remains leaked.	**FAIL**	Missing delete[].